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Optimizing adaptive modulation technique using standard propagation model for enhanced wireless communication channels

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Recent progress in the domain of wireless communication systems has brought to light emerging trends that underscore the evolution of modern communication paradigms. Nonetheless, the performance of such systems remains constrained by persistent technical limitations, including packet loss, bandwidth scarcity, and suboptimal spectral efficiency, all of which necessitate rigorous analytical investigations. To address these constraints, the present study leverages an adaptive modulation technique tailored to enhance communication efficiency. This technique is systematically evaluated considering various parameters, including the signal-to-interference-plus-noise ratio (SINR), link distance, and the Standard Propagation Model (SPM), across canonical wireless channel environments, such as Additive White Gaussian Noise (AWGN), Rayleigh, and Rician fading models. The primary objective is to minimize the Bit Error Rate (BER), maximize system throughput, and ensure efficient utilization of available bandwidth by strategically optimizing the SPM parameters for long-range cellular communication links. Simulation outcomes substantiate that the proposed SPM-integrated framework achieves a target BER = 10^{-2} and exhibits enhanced spectral efficiency for 64-QAM modulation across a 6 km cell radius, thereby outperforming the conventional non-SPM approach. Notably, the proposed technique achieves SINR improvements of 40%, 43%, and 41% under AWGN, Rayleigh, and Rician conditions, respectively. To ensure optimal configuration, an advanced optimization algorithm is employed to dynamically select the most effective SPM parameters, enabling robust performance across varying channel conditions. Moreover, empirical analysis confirms that the SPM-enhanced system architecture facilitates the stable deployment of 64-QAM across all tested channel models, consistently yielding superior SINR and reduced BER relative to baseline techniques that lack SPM integration.

Keywords Adaptive modulation, M-ary Phase-Shift keying, M-ary quadrature amplitude modulation, Signal-to-Interference-Plus-Noise ratio, Bit error rate, Different communication channels, Standard propagation model

Information and Communications Technology (ICT) is essential for the growth of a nation¹ and digitization, covering fields like information, communication, and telecommunication technologies. Within this context, wireless telecommunication networks signify a basis of ICT infrastructure, playing an essential role in enabling the unprecedented coverage and continuous connectivity that moves social transformation, thereby exerting a profound influence on various aspects of daily life through emerging technologies². Its adoption fosters economic development, enhances employment and skill levels, and promotes transparent, accountable, and citizen-focused governance. ICT, especially mobile communication technology, can improve life and livelihood, particularly in rural areas. Developed countries are leveraging advanced network technologies like 5G for

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higher data rates and connectivity. However, economically challenged countries like Pakistan, India, Kenya, Afghanistan, Bangladesh, Sri Lanka, and Somalia are lagging in technology adoption, still reliant on 3G and 4G networks. Issues such as distant base stations (BS's), low signal strength, and network interference from multiple operators sharing equipment result in unreliable connections and reduced internet speed, especially in rural and urban environments. Unlike developed nations, these underprivileged countries still use multiple-generation communication systems, like 2G, 3G, and 4G simultaneously. While 2G and 3G are primarily used for voice calls and SMS, 4G is used for internet browsing as well. Poor cellular signal is a common issue due to factors such as the long distance between BS's, a large number of obstacles, and physical obstructions. As a result, the coverage area of the cellular communication system is affected, leading to frequent call drops and limited network accessibility. Figure 1 shows multi-generation communication systems like 2G-4G in the urban and suburban complex environments in non-developed areas.

The increasing need for high data rates and quality of service (QoS) in communication systems is propelling notable technological progresses on a global scale³. The upcoming generation of wireless systems must be able to provide high data rates over extensive distances without the need for extra bandwidth. Furthermore, they should also possess enhanced mobility, strong links, superior voice quality, high transmission power and coverage, and exceptional QoS⁴. To meet the increasing data rates and growing usage of information networks, innovative technologies are being implemented to enhance the performance of wireless communication networks⁵. These systems are required to function efficiently in a variety of transmission environments, such as urban, suburban, and rural areas, as well as in different multipath and time-varying fading channels. In situations where the transmitter and receiver are positioned far apart without a direct line of sight (LOS), communication relies on multipath channels, which can potentially lead to data loss⁶. The large-scale fading describes changes in signal strength over long distances (hundreds or thousands of meters) between transmitter and receiver, including path loss and shadowing⁷. The performance of communication systems can be negatively impacted by channel variations⁸. To decrease the bit error rate (BER) within these constraints, specific modulation and coding schemes are applied, as increasing transmit power or bandwidth may not be feasible due to infrastructure limitations. Different mobile generations (1G through 5G) employ unique technologies and modulation schemes. The term 'mobile wireless generation' refers to changes in system nature, frequency technology, latency, speed, data capacity, and more⁹. The technology, modulation scheme, channel conditions, and propagation models used in mobile cellular communication affect spectral efficiencies and BER.

The Standard Propagation Model (SPM) is a commonly utilized mathematical framework in wireless mobile communication for measuring path loss, or the reduction in the strength of a signal, as it passes through different environments. It enables engineers to measure signal strength at a specific distance from the transmitter, considering both the physical environment and the features of the signal's propagation. SPM is empirical, depending on extensive data collection for a given situation, allowing for realistic predictions of a link's behaviour¹⁰. This model is particularly effective in predictive Large-scale fading over distances ranging from 1 to 20 km. Furthermore, Signal coverage prediction in a cellular network can be improved using calibrated SPM, which can be tuned for precision, especially in built-up and urban environments.

Standardized propagation models have been developed over the past few decades to tackle the increasing complexity of wireless communication systems and provide a solid foundation for these systems¹¹. The SPM is characterized by some important features that improve its flexibility and accuracy in different communication environments. Its environmental flexibility enables successful use across various setting, such as urban, suburban,

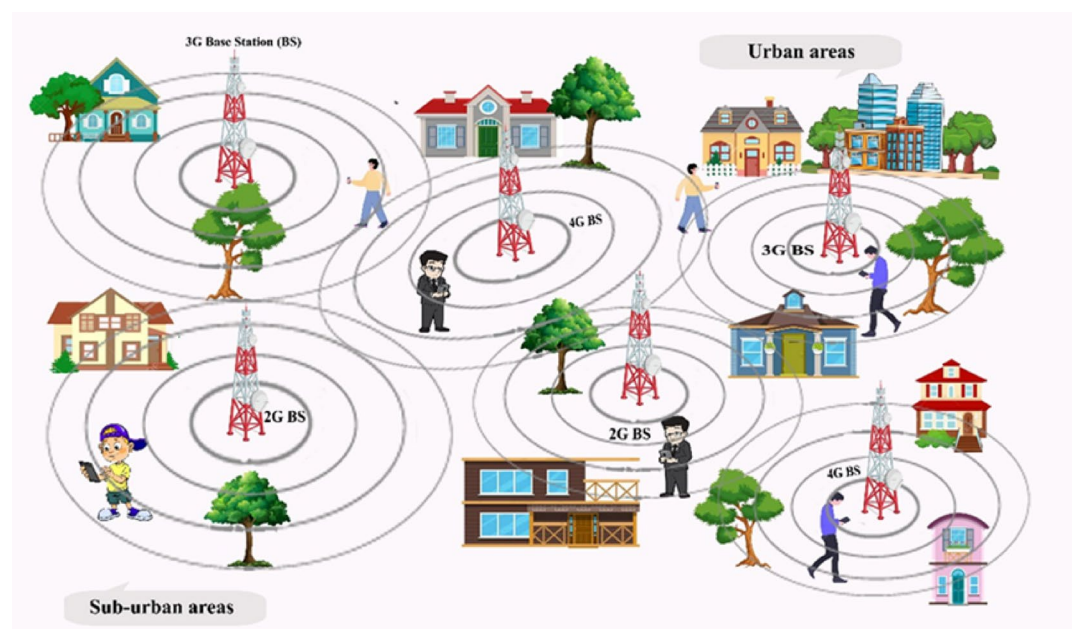


Fig. 1. A multi-generation communication systems in an urban and suburban of non-developed areas.

rural, and indoor environments, thus making it appropriate for city centers, highways, forests, and buildings. For example, in densely built metropolitan regions, signal loss is often more due to constraints such as buildings, whereas rural areas have less path loss owing to fewer obstacles. The model incorporates a path loss exponent that represents how signal strength diminishes with distance, with the value of the exponent varying depending on the environment. In free-space environments, the exponent may be 2, however in metropolitan regions, it may raise to 3 or because of extra obstacles and signal scattering. The value of path loss exponent is highly dependent on the carrier frequency and the LOS state between the BS and mobile user equipment¹². Moreover, the SPM addresses multi-path fading, a phenomenon where signals take multiple routes to the receiver, causing interference, specially in urban environments with reflective surfaces. This also addresses shadowing, defined as signal attenuation due to substantial obstacles like buildings or mountains, efficiently considering real-world complexities that basic models usually overlook, which presume a direct LOS between the transmitter and receiver. These features make the SPM a more reliable and comprehensive model for predicting signal behavior across diverse environments.

Moreover, the SPM is better than traditional models in several important ways, such as being more flexible, more accurate, and better at considering real-world problems. Accurate estimation of path loss is essential in cellular network design^{13,14}. The SPM is very flexible and can accurately predict path loss in a wide range of settings, including rural, urban, suburban, and indoor environments. This is different from traditional models like the path loss model, the Okumura-Hata model, the Cost 231-Hata model, the ECC-33 model, and the Egli path loss model, which usually only focus on certain types of environments, like urban areas or free-space scenarios. It adapts dynamically to variables like topography, vegetation, and obstructions. The SPM considers complex real-world effects like multi-path fading and shadowing, rendering it particularly appropriate for urban and cluttered environments. Conventional models tend to oversimplify these conditions and often do not effectively predict signal behavior in less controlled or dynamic environments.

Furthermore, the SPM integrates environmental elements with real-world data, yielding a precise representation of signal strength in practical contexts. Conventional models rely on theoretical assumptions or limited empirical data from specific regions, thereby reducing their accuracy and applicability across varied conditions. Thus, whereas conventional models may suffice in simple or more controlled setting, the SPM distinguishes itself by its superior accuracy and wider application, due to its reliance on actual data owing to its reliance on empirical data and its ability to adjust to diverse situations.

Various issues, including noise, interference, propagation loss, bandwidth limitation, and multipath fading, affect communication systems^{15,16}. Digital modulation systems effectively address these problems and outperform analog systems. They are less susceptible to noise and interference and offer improved bandwidth efficiency¹⁷. There is an inherent trade-off between operating bandwidth and noise performance, commonly referred to as the matching-bandwidth noise figure limitation¹⁸. Digital modulation provides advantages such as increased information transmission capacity, data security, higher transmission quality, rapid system availability, and allocation of RF spectrum for additional services^{19,20}. The optimal modulation scheme depends on factors like signal-to-noise ratio (SNR), QoS, BER, power efficiency, and cost^{21,22}. The error probability of different modulation system is assessed by analyzing its performance in the existence of an Additive White Gaussian Noise (AWGN) channel²².

Hanzo and Torrance have introduced an adaptive modulation system designed for wireless communication, aiming to achieve low BER and high QoS. This system adapts modulation schemes based on the prevailing channel conditions¹⁷ to effectively utilize bandwidth, enhance data rates, and reduce BER. It manipulates wireless channel information, providing improved performance over fading channels²³, compared to other systems that don't use this knowledge. Adaptive modulation is an appealing technology for enhancing transmission efficiency over wireless fading channels while ensuring reliability²⁴ and balances BER with spectral efficiency^{8,9}. The system adapts transmission parameters such as modulation power, coding rate, and mode based on channel conditions²⁵. This study introduces an entirely novel adaptive modulation method utilizing SPM, which is founded on the signal-to-interference-plus-noise ratio (SINR) and distance. This approach improves high-speed data transmission over long distances and in densely populated areas, particularly in underdeveloped countries. SPM enhances signal coverage in built-up and urban areas by facilitating high-frequency signal transmission over large distances (1–20 km). This technique optimizes SPM parameters to reduce the BER and boost SINR, thereby improving communication services in underprivileged regions. The use of SINR and distance threshold-based adaptive modulation in different fading channels is relatively unexplored, and no similar research with these constraints and system model exists to the author's knowledge. The study's primary contributions are outlined below.

- For adaptive modulation with or without using SPM, a distance-based threshold is proposed to facilitate a more comprehensive comparison between the proposed methods and baseline approaches. This is with the aim of achieving the desired target BER in the long-range communication cellular system.
- This work aims to examine the impacts of co-channel interference (CCI) from BS's on users and total modulation systems. Combined interference considering all BS's in the cluster is modeled as the SINR ratio. In this case, CCI is received together with the desired signal from the transmitter and delivered to the mobile user. The transmitting base station (BS) that transmits the signal to the user also adjusts the modulation scheme through SINR in different wireless channels with the aim of improving BER and throughput.
- Different modulation schemes and their BER performances are evaluated and scrutinized using the proposed technique (with or without SPM) in both AWGN and realistic channel scenarios like Rician and Rayleigh fading channels.
- The results show that our developed technique using SPM exhibits excellent performance, compared to the technique without using SPM, in terms of BER and SINR at long-range distances.

The rest of the article is organized as follows: Related work is presented in Sect. 2, the methodology is described in Sect. 3, the simulation results are discussed in Sect. 4, and conclusions are given in Sect. 5.

Related work

The Adaptive Modulation and Coding (AMC) algorithm selects suitable code rates and modulation orders by using the measured SNR and BER. When the SNR is high and the BER is relatively low, it utilizes higher modulation orders and coding rates, such as 256-QAM with a 3/4 code rate, to improve the efficiency of using the available frequency spectrum. On the other hand, when faced with difficult channel conditions, it uses less complex modulation orders and coding rates, such as BPSK with a 1/4 code rate, to maintain a reliable connection²⁶. However, in cognitive scenarios, the analytical method differs due to potential transmission delays and packet failures from collisions. The integration of large packet transmission with AMC is facilitated through Enlightened Data Transport, which offers multiple transmission slots based on secondary channel behavior²⁷. This study presents and examines a spectral-efficient adaptive modulation strategy for heterogeneous Two-Way Relay Networks that utilize symbol-based networks. The proposed method allows transceivers to employ several modulation schemes at varying bit rates, while adapting these parameters to the specific conditions of the channel. The proposed rate adaption technique effectively reduces the transmission of errors from the relay to transceivers by ensuring that the entire BER is limited according to QoS requirements. Both transceivers utilize identical modulation methods²⁸.

Rogozhnikov EV et al. proposed an adaptive modulation technique within the framework of orthogonal frequency-division multiplexing (OFDM) communication systems, which would help reduce bit errors, particularly under harsh conditions of low signal-to-noise ratios and interference. The proposed technique utilizes pilot symbols for data link estimation and successively calculates the maximum error vector magnitude. As compared to the existing techniques, the technique reduces the bit error rate by 15% and improves spectral efficiency by a factor of 1.5²⁹. Wang K et al. addressed the uncertainty in BER and the performance of throughput inherent in uplink non-orthogonal multiple access systems. Given the closed-form BER expressions, the authors develop an asymmetric adaptive modulation framework and an algorithm for performance improvement. The proposed algorithm outperforms the existing benchmarks in system sum throughput³⁰. Dilshad AA et al. proposed an adaptive system that integrates the principles of OFDM and Multi-Carrier Code Division Multiple Access (MC-CDMA) to enhance data rates in wireless communication. They have analyzed the performance of OFDM and MC-CDMA separately over Rayleigh and AWGN channels for BER. They have designed an adaptive system that switches between OFDM and MC-CDMA based on the SNR of the channel³¹. Yang F et al. introduced a new approach to adaptive modulation in smart systems, referred to as non-data-aided error vector adaptive modulation, which utilizes a non-data-aided error vector for channel quality assessment to adapt the transmission rates. They analyze its performance using a finite-state Markov chain model and simulations, which show that it has improved spectral efficiency and reduced packet loss compared to traditional adaptive modulation techniques³². Aljohani AJ and Ng SX proposed a novel bandwidth-efficient communication scheme: Distributed Joint Turbo Trellis-Coded Modulation aided either by dynamic network coding (DNC) or adaptive DNC. These schemes aim to ensure minimum transmission energy by exploiting the correlation between transmitted signals and network coding, with the aid of channel coding, thereby ensuring ultra-reliable communication over noisy channels³³.

Saatchi NS et al. addressed the challenge of ultra-reliable and low-latency communication for industrial IoT applications on 5G networks. The authors propose data transmission optimization by dynamically selecting the best communication settings—such as numerology, mini-slot size, and others—on each attempt, depending on the connection quality and remaining time. To solve this optimization problem, the suggested approach is model-based reinforcement learning³⁴. Polus R and D'Amours C examined the ergodic capacity of unmanned aerial vehicle-to-ground wireless channels, concentrating on different power adaptation approaches. They present closed-form solutions for channel capacity and effective capacity across various strategies, including optimal rate adaptation with fixed power, optimal power and rate adaptation, channel inversion with fixed rate, and truncated channel inversion with fixed rate, and validate these through Monte Carlo simulations³⁵. Tian T et al. explored adaptive bit and power allocation using beamforming to enhance the BER within dual-function radar-communication systems, setting up a BER minimization problem that addresses radar and resource limitations and designing an alternating optimization algorithm to resolve it effectively. Simulation results showed improved communication performance compared to current methods³⁶. Mahmud M et al. introduced two groundbreaking technologies for air-to-underwater communication utilizing nonlinear optoacoustics: optical focusing-based adaptive modulation (OFAM)-1D and OFAM-3D. These approaches utilize dynamic optical focusing to control an underwater sound source, resulting in significantly higher data rates and energy efficiency compared to traditional methods. Furthermore, the authors emphasize the application of machine learning algorithms to improve signal demodulation resilience in this scenario³⁷. Adaptive modulation is gaining traction in millimeter-wave (mmWave) communication systems, which operate at substantially higher frequencies, typically in the 30–300 GHz range, compared to regular wireless networks. These systems have the potential for extraordinarily fast data transport, but they are also susceptible to propagation problems like as air absorption and physical impediments³⁸. Wang Lei et al. have also considered adaptive modulation in indoor infrared wireless communication setups. For these systems, adaptive modulation can help offset multipath propagation issues, thereby enhancing overall performance³⁹. Smida Besma et al. summarized the literature review, underscoring the increasing significance of adaptive modulation techniques in wireless communication, especially for high-speed data transfer and managing channel-related limitations⁴⁰.

Several modulation approaches enable the transmission of a greater number of bits per symbol, resulting in higher data transfer rates. Modulation methods such as 64-Quadrature Amplitude Modulation (64-QAM) and 16-QAM require a greater SNR in order to effectively minimize the presence of noise and interference. Wireless

systems were developed to adjust the modulation order according to the channel characteristics⁴¹. Dynamic adaptive modulation techniques, along with forward error correction coding, are employed for minimizing the BER in multiple-input multiple-output OFDM systems. The choice of high modulation schemes results in high data rates but with reduced immunity to noise, as demonstrated by the BER versus SNR analysis⁴². The author has demonstrated how modulation and coding can be used to minimize interference. Few efforts have been made to address interference issues in dense networks through the use of switching mechanisms⁴³. N-MSK modulation schemes are preferred over N-QAM adaptive modulation in scenarios with limited power sources, as the former is less affected by non-linear transmitter amplification. OFDM systems efficiently utilize the available spectrum by dividing the channel into sub-channels that experience flat fading⁴⁴. To enhance spectral efficiency, advanced modulation schemes, such as high-order QAM, are utilized. Additionally, coding techniques with faster code rates are applied to further boost wireless channel throughput. However, QAM with a lower order is more prone to demodulation errors, whereas a coding scheme with a higher coding rate offers limited error correction capabilities. A trade-off is crucial because there is a conflict between the QoS and the spectral efficiency performance of the wireless channel. The trade-off can be accomplished by carefully choosing the modulation and coding scheme. The AMC approach is commonly employed in current wireless systems to enhance data throughput and fulfill QoS requirements in the presence of a time-varying channel²⁵.

Methodology

The attenuation, interference, and distortion caused by multipath channel impairment adversely disrupt the transmission of various modulated signals from a transmitter to a receiver, including binary phase shift keying (BPSK), quadrature phase shift keying (QPSK), 16-QAM, and 64-QAM. The proposed approach using SPM, illustrated in Fig. 2, employs a modulator selector acting as a feedback loop for the transmitter. It gathers data on distance and SINR across various channel conditions like AWGN, Rayleigh, and Rician fading channels, as determined by the channel estimator, which is part of the receiver. This data is then relayed to the transmitter by the modulator selector, which adapts modulation schemes according to the prevailing conditions in both AWGN and multipath fading channels, including Rayleigh and Rician channels. Figure 2 depicts the adaptive modulation model employing SPM. The adaptive modulation technique used in this study is a discrete one, which switches between (BPSK, QPSK, 16-QAM, and 64-QAM) according to channel conditions (Such as SINR and distance) in AWGN, Rayleigh, and Rician Fading channels.

The mathematical equation for SINR of various modulated signals is shown under

$$\text{SINR} = \frac{P_{\text{Signal}}}{I + P_{\text{Noise}}} \quad (1)$$

The signal power from the primary BS is denoted by P_{Signal} , while I represents the combined CCI from six BS's affecting the mobile user who is at a 3 km distance and at a 35° angle from the main BS. P_{Noise} signifies the power of the channel noise. Equation (2) displays the interference power between the user and the adjacent mobile BS.

$$P = P_0 \left(\frac{d_0}{d} \right)^n \quad (2)$$

In free space, the constant path loss exponent denoted by n and is equal to 2 and d_0 represents a reference distance equal to 100 m because it is 1–100 m in the outdoor environment. P_0 denotes the power received at the reference distance d_0 as illustrated in Eq. (3):

$$P_0 = \frac{\lambda^2 P_t G_T G_R}{(4\pi)^2 d_0^2} \quad (3)$$

The signal wavelength is symbolized by λ , while the transmitted power is indicated by P_t . The antenna gain for the transmitter and receiver are represented by G_T and G_R respectively. To compute the distance d between a user and interfering mobile BS's, one must first establish D , which is the distance between the principal BS and the co-channel mobile BS's, in the following manner:

$$D = R\sqrt{3N} \quad (4)$$

R denotes the radius of the BS, assumed to be 6 km, while N signifies the cluster size, which is 7 in our case. The six co-channel BS's form different angles including 30°, 90°, 150°, 210°, 270° and 330° with the main mobile BS. The following equations can be used to calculate the coordinates of each of the six co-channel BS's:

$$\mathbf{X} = D \cos(\theta) \quad (5)$$

$$\mathbf{Y} = D \sin(\theta) \quad (6)$$

In the equation shown above, D represents the distance, and θ indicates the angle between the primary and co-channel mobile BS. Similarly, we can calculate the coordinates of the six neighbouring BS's. The above same equation may also be used to get the coordinates of the mobile user by taking $D' = 3$ km and $\theta' = 35^\circ$. To determine the distance between the mobile user and each neighboring BS, the Euclidean distance d_i between (X_i, Y_i) and $(X_{\text{Mobile user}}, Y_{\text{Mobile user}})$ is shown by the following equation:

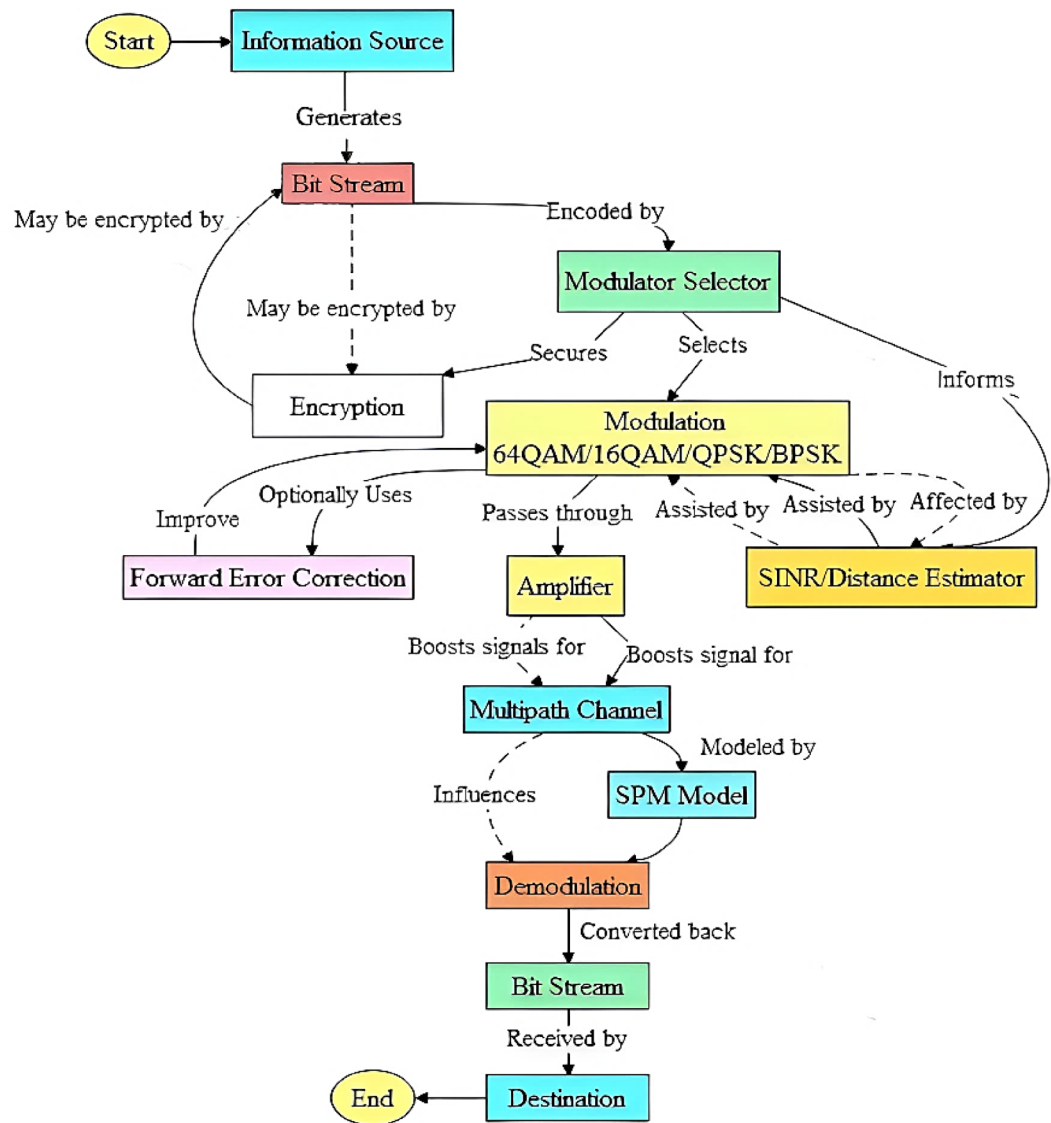


Fig. 2. Flow chart of the proposed dynamic adaptive modulation system deploying SPM.

$$d_i = \sqrt{(X_i - X_{\text{Mobile user}})^2 + (Y_i - Y_{\text{Mobile user}})^2} \quad (7)$$

Where X_i , Y_i and $X_{\text{Mobile user}}$, $Y_{\text{Mobile user}}$ are the X and Y coordinates of the i th neighbouring mobile BS's and mobile user location respectively in the above equation. After computing d_i , the interference power from the surrounding neighbouring mobile BS's has been measured. In our specific case, there are six neighbouring co-channel BS's so the interference power $P(I)$ is the accumulation of interference power from all six co-channel BS's as described:

$$P(I) = \sum_{i=1}^6 P(I)_i \quad (8)$$

Substituting Eq. 8 into Eq. 1, the SINR formula becomes

$$\text{SINR} = \frac{P_{\text{Signal}}}{\sum_{i=1}^6 P(I)_i + P_{\text{Noise}}} \quad (9)$$

Where, $\sum_{i=1}^6 P(I)_i$ represents the accumulation of interference power from the six neighbouring co-channel BS's operating on the same frequency channel. This CCI is a primary factor degrading the signal quality of the primary BS, thereby affecting the mobile user actively communicating with it within its coverage area.

For modeling SINR with CCI, we calculate or measure the desired signal power and the accumulated interference power from all CCI sources. These values are then substituted into the SINR formula, as given in Eq. 9.

The equation shown below calculates the probability of error for BPSK and QPSK in the AWGN channel.

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \quad (10)$$

Hence, E_b represents energy per bit, N_0 represents noise power per bit and $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt$. The equation below represents the error probability of an M-QAM-modulated signal in an AWGN channel.

$$P_b = 4 \frac{\sqrt{M}-1}{\sqrt{M}} Q\left(\sqrt{\frac{3}{M-1} \frac{XE_b}{N_0}}\right) - 4 \left(\frac{\sqrt{M}-1}{\sqrt{M}}\right)^2 Q^2\left(\sqrt{\frac{3}{M-1} \frac{XE_b}{N_0}}\right) \quad (11)$$

$X = \log_2 M$ represents the bits per symbol, Where M indicates the modulation constellation size. The capacity of the AWGN channel denoted by C_{AWGN} is defined as the maximum of mutual information $I(X; Y)$ concerning the probability density function f_X of the channel input X and computed as,

$$C_{AWGN} = \max_{f_X} I(X; Y) = \frac{1}{2} \log_2 (1 + SNR) \quad (12)$$

The following Eq. (13) may be used to compute the probability of error of MPSK in fading channel environment is :

$$P_b = \frac{1}{\pi} \int_0^{(M-1)\pi/M} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{\sin^2(\pi/M)}{\sin^2\theta} \right) d\theta \quad (13)$$

The probability of error for MQAM in a fading channel is depicted as follows

$$P_b = \frac{4}{\pi} \left(1 - \frac{1}{\sqrt{M}}\right) \int_0^{\pi/2} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{3/(2(M-1))}{\sin^2\theta} \right) d\theta - \frac{4}{\pi} \left(1 - \frac{1}{\sqrt{M}}\right) \int_0^{\pi/4} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{3/(2(M-1))}{\sin^2\theta} \right) d\theta \quad (14)$$

The probability of error for MPSK and MQAM in Rayleigh and Rician channels is identical, with the only distinction being the value of M_{γ_l} . In Rayleigh, M_{γ_l} is expressed in Eq. (15), while in Rician channels, it is provided in Eq. (16):

$$M_{\gamma_l}(s) = \frac{1}{1 - s\gamma_l} \quad (15)$$

$$M_{\gamma_l}(s) = \frac{1 + N}{1 + N - s\gamma_l} e^{\left[\frac{Ns\gamma_l}{(1+N)-s\gamma_l} \right]} \quad (16)$$

Hence γ_l denotes the SNR per symbol per branch and it is equal to $(\Omega_l \frac{XE_b}{N_0})/L$, Ω_l is the power of the fading amplitude r and it is equal to $\Omega = E[r^2]$ and L denotes the number of diversity branches. N denotes the ratio of energy in the specular component to the energy in the diffuse component. $\bar{\gamma}_l = \gamma_l$, for identically distributed diversity.

The Rayleigh distribution is given below,

$$P(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{\sigma^2}\right) \quad 0 \leq r \leq \infty \quad (17)$$

The rms value of the received signal is represented by σ , where instantaneous power is denoted as $\frac{r^2}{2}$ and local average power of the received signal before detection is indicated by σ^2 .

The probability density function of Rician fading channel is given below,

$$f(r|s, \sigma) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + s^2}{2\sigma^2}\right) I_0\left(\frac{sr}{\sigma^2}\right) \quad \text{for } s \geq 0, r \geq 0 \quad (18)$$

Where $I_0(\dots)$ is the modified Bessel function of the first kind with order zero. $2\sigma^2$ is the average power in the non-LOS multipath components, s^2 is the average power in the LOS component and $\frac{r^2}{2}$ is the instantaneous

power. The mobile user moves within a 6 km coverage area of the BS at velocity V to analyze the relationship between SINR, BER, and distance, with a signal frequency of f_c . The equation below illustrates the received signal power.

$$P(r) = P_t + G_r + G_t - L_c - \text{SPM}_{(\text{Path loss})} \quad (19)$$

In the above Eq. (19), L_c represents the antenna signal transmission loss, P_t denotes the transmitted signal strength and $\text{SPM}_{(\text{Path loss})}$ characterizes the standard propagation model for path attenuation. The $\text{SPM}_{(\text{Path loss})}$ can be expressed in Eq. (20).

$$\begin{aligned} \text{SPM}_{(\text{Path loss})} = & K_1 + K_2 \log(d) + K_3 \log(H_{\text{Txeff}}) + K_4 \times \text{Diffraction loss} \\ & + K_5 \log(d) \times \log(H_{\text{Txeff}}) + K_6 (H_{\text{Rxeff}}) + K_7 \times \log(H_{\text{Rxeff}}) + K_{\text{clutter}} \times f_{\text{clutter}} \\ & + K_{\text{hill,LOS}} \end{aligned} \quad (20)$$

A list of all key SPM parameters used in the above mathematical Eq. 20 of SPM is given below.

K_1^{45} = Constant offset (dB).

K_2 = Multiplying factor for $\log(d)$.

d = Distance between Transmitter and Receiver (meters)

K_3 = Multiplying factor for $\log(H_{\text{Txeff}})$.

H_{Txeff} = Effective height of the Transmitter antenna (meters).

K_4 = Multiplying factor for diffraction calculation.

Diffraction Loss = Losses due to diffraction over an obstructed path (dB).

K_5 = Multiplying factor for $\log(d) \times \log(H_{\text{Txeff}})$.

K_6 = Multiplying factor for H_{Rxeff} .

K_7 = Multiplying factor for $\log(H_{\text{Rxeff}})$.

H_{Rxeff} = Effective height of the Receiver mobile antenna (dB).

K_{clutter} = Multiplying factor for $f(\text{clutter})$.

$f(\text{clutter})$ = Average of weighted due to clutter.

$K_{\text{hill,LOS}}$ = Corrective factor for hilly regions.

The equation provided below can be utilized to compute the SINR in dB at distance d .

$$\text{SINR}(\text{dB}) = P_{(r)}\text{dB} - P_{(\text{Noise})}\text{dB} - P(I)\text{dB} \quad (21)$$

Lastly, the formula for determining SINR is presented as follows

$$\text{SINR} = P_t + G_r + G_t - L_c - \left(\begin{aligned} & K_1 + K_2 \log(d) + K_3 \log(H_{\text{Txeff}}) + K_4 \times \text{Diffraction loss} \\ & + K_5 \log(d) \times \log(H_{\text{Txeff}}) + K_6 (H_{\text{Rxeff}}) + K_7 \times \log(H_{\text{Rxeff}}) + K_{\text{clutter}} \times f_{\text{clutter}} \\ & + K_{\text{hill,LOS}} \end{aligned} \right) - P_{(\text{Noise})} - P(I) \quad (22)$$

The proposed Dynamic Adaptive modulation approach deploying SPM is given as Algorithm 1.

The Proposed Dynamic Adaptive Modulation Algorithm

Initialization	Initialize parameters: modulation schemes M_i , coding rates R_i , SINR thresholds γ_{th} , distance threshold d_{th} , and detailed propagation model parameters P . Set channel model parameters for AWGN, Rayleigh, and Rician fading channels including antenna gains G_T , G_R , transmitted power P_t and environment characteristics.
Packet Transmission Loop	Loop over each packet transmission, adjusting for channel type (AWGN, Rayleigh, Rician) and corresponding parameters.
SINR and Distance Measurement	Measure received signal power (P_r), Interference (I) and Noise power (N). Calculate SINR using $SINR = \frac{P_r}{I+N}$. Estimate d using $d = SPM(d_{th}) + \epsilon$, Where SPM is the Standard Propagation Model with enhanced models considering antenna characteristics and environmental factors and ϵ is random deviation.
Modulation and Coding Determination	Calculate the BER from equation such as $P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$ for BPSK/QPSK and the equation for MQAM in AWGN channel is $P_b = 4 \frac{\sqrt{M}-1}{\sqrt{M}} Q\left(\sqrt{\frac{3}{M-1} \frac{XE_b}{N_0}}\right) - 4 \left(\frac{\sqrt{M}-1}{\sqrt{M}}\right)^2 Q^2\left(\sqrt{\frac{3}{M-1} \frac{XE_b}{N_0}}\right).$ The probability of error for MPSK in fading channel is $P_b = \frac{1}{\pi} \int_0^{(M-1)\pi/M} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{\sin^2(\pi/M)}{\sin^2\theta} \right) d\theta,$ while the probability of error for MQAM in fading channel is $P_b = \frac{4}{\pi} \left(1 - \frac{1}{\sqrt{M}}\right) \int_0^{\pi/2} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{3/(2(M-1))}{\sin^2\theta} \right) d\theta - \frac{4}{\pi} \left(1 - \frac{1}{\sqrt{M}}\right)^2 \int_0^{\pi/4} \prod_{l=1}^L M_{\gamma_l} \left(-\frac{3/(2(M-1))}{\sin^2\theta} \right) d\theta$
Transmission	Determine Modulation order (m) and coding (r) based on SINR and distance. Modulation order: $m = \operatorname{argmax}_M \{SNR - \gamma_{th}\}$ Code Rate: $r = \operatorname{argmax}_R \{SNR - \gamma_{th}\}$
Receive and decode	Transmit the packet using the selected modulation and coding scheme (m , r) Receive the transmitted packet and decode using the same modulation and coding scheme
Feedback Adaptation	Collect feedback: ACK/NACK, received signal strength and packet error rate Adjust parameters based on feedback. Update SINR Threshold: $\gamma_{th} = f(\gamma_{th}, \text{feedback})$ Update Distance Threshold: $d_{th} = f(d_{th}, \text{feedback})$ Update Modulation and Coding Parameters: Adjust M and R based on feedback and error probability calculation.
End for	Continue looping over packet transmissions, with dynamic adjustments for optimization.

Algorithm 1 Dynamic Adaptive Modulation Approach deploying SPM

Inclusion of forward error correction (FEC) in communication systems

Forward Error Correction (FEC) is an important mechanism used in communication systems that, in an effort to increase the reliability of data transmission, especially in unreliable or noisy environments. FEC enables the correction of errors on the received data without retransmission by adding redundancy to the data which is sent. It is an important feature of the communication system because it significantly enhances the efficiency

and performance of the communication scheme, especially where retransmission is expensive or impossible to justify, such as in real-time communication or various satellite systems. In the present study, we examine the role of FEC in conjunction with various equalization schemes, including LE, DFE, adaptive equalizers, and SPM. The inclusion of FEC is especially pertinent in environments where signal degradation due to noise, fading, and multipath effects is common. Using FEC, we would strengthen the communication system, giving us an added degree of protection against errors. We shall take into consideration some of the typical fashions of FEC schemes that are:

Block codes

The block codes work by breaking data into fixed-size blocks and adding parity bits, which enables the receiver to recover the errors. Hamming Codes, BCH Codes, and Reed-Solomon Codes are some examples of block codes. These codes easily counteract mistakes in structured information and are widely employed within communication outlets, such as satellite or deep-space communications.

Convolutional codes

Contrary to the aforementioned codes, these codes encode data that is grouped into a continuous stream, rather than fixed-size blocks. They are typically applied in systems where real-time transmission of data is required, performing well in combination with methods such as Viterbi decoding to mitigate errors. Such convolutional codes are generally utilized within mobile communication systems, such as GSM and CDMA.

Turbo codes

Turbo coding is the combination of two or more convolutional codes interleaved by some amount. Turbo codes achieve good absolute performance close to that of the Shannon limit, i.e., close to error-free coding. These codes are commonly used in wireless communication, such as 3G and 4G networks, where error rates and high data rates are crucial.

Low-density parity-check codes (LDPC)

LDPC codes are a new type of code that can approach the theoretical performance of error correction. They are common in more recent communication standards, including Wi-Fi, DVB-S2 (satellite communication), and 5G networks, where they offer high error-correction rates at low complexity.

FEC and its effects on performance

The effect of FEC on overall system performance, in combination with SPM and the number of equalization techniques, has been evaluated in this study. The following factors are taken into consideration:

Reducing error rate

FEC schemes, especially Turbo Code and LDPC, are likely to significantly reduce the rate of BER as well as the rate of symbol error (SER) in highly hostile interference and multipath environments.

Mixing with equalizers

The performance of FEC will be tested along with various equalizers (LE, DFE, and adaptive equalizers). We have noticed that when used in conjunction with equalizers, FEC has further enhanced the system’s reliability, particularly in combating errors caused by multipath fading and inter-symbol interference (ISI).

Channel capacity

The functioning of FEC when it comes to maximizing the available channel capacity is by lessening the number of retransmissions. This is especially true for high-speed communication systems, where the limiting factor is bandwidth.

Incorporating the effective technique of FEC into our review, we have endeavored to provide a comprehensive assessment of the value of error correction measures in enhancing the effectiveness and reliability of a communication system, with an emphasis on experimentation in a challenging channel condition using equalization methods. Table 1, given below, displays a summary of Common Forward Error Correction (FEC) Schemes and their applications.

Experimental results and discussions

This section of the article discusses the performance evaluation of different modulation schemes, including 64-QAM, 16-QAM, QPSK, and BPSK, utilized in an adaptive modulation technique with SPM, concerning SINR versus Distance and BER versus Distance. The most widely used telecommunications channels for observing

FEC Scheme	Key Characteristics	Common Applications
Block Codes	Encode data in fixed-size blocks with added parity bits.	Satellite communications, storage systems, and CD/DVD media.
Convolutional Codes	Encode data as a continuous stream with memory of previous inputs.	Mobile communications (GSM, CDMA), satellite systems.
Turbo Codes	A combination of two or more convolutional codes with interleaving.	Wireless communications (3 G, 4G), deep-space communications.
LDPC Codes	High-performance codes approaching the Shannon limit error correction.	Wi-Fi, DVB-S2, 5G networks.

Table 1. Summary of common forward error correction (FEC) schemes and their Applications.

the effects of interference and fading are AWGN, Rayleigh, and Rician channels⁴⁶. We analyze CCI in our study, which is a specific form of inter-cell interference. It arises as a result of the overlap of signals from neighboring cells that are using the same frequency, a key characteristic of inter-cell interference. We deploy an adaptive modulation approach with/without using SPM instead of using dynamic frequency reuse or power control to mitigate the effect of CCI from neighboring BS's and improve network performance. This analysis takes into account the interference from six co-channel mobile BS's in Rayleigh, Rician, and AWGN channels. The performance of the proposed technique using SPM is compared to that of an adaptive modulation technique without SPM. The coexistence of mobile BS's interference affects the targeted signal, which then undergoes Rician, Rayleigh, and AWGN channels during transmission. The different parameters of SPM are optimized to achieve a target BER = 10^{-2} with a high SINR for the aforementioned channels, as defined in Table 2.

The SPM parameters K1 to K7 are adaptable based on various propagation conditions such as terrain profile, diffraction mechanisms, clutter class morphology, and antenna height. Calibration of these parameters aims to reduce the discrepancy between path loss predictions and actual measurements. This calibration process involves a computational approach that adjusts the propagation model's formula, yielding results that closely mirror real measurements. The calibrated SPM enhances model fit, improving metrics such as Mean Absolute Error, Root Mean Square Error, and Standard Deviation, compared to uncalibrated parameters. Additionally, it adjusts for path loss caused by altitude differences between Unmanned Aerial Systems and cell towers, improving signal coverage prediction accuracy within a 1–20 km range.

The below Fig. 3 depicts the distance and angle between the primary BS and six co-channel mobile BSs. It also shows the distance between the mobile user and the co-channel BS's. The mobile user is located 3 km away from the primary BS and is positioned at an angle of 35° relative to the primary BS. The main BS has a coverage area of 6 km and is experiencing interference from six surrounding co-channel BS's. Each interfering BS is 24.50 km, 25.88 km, 28.87 km, 30.46 km, 29.31 km and 26.35 km away from the targeted mobile user, as shown in Fig. 3. Each interfering BS has a distinct level of interference on the mobile user. Table 3 displays the simulation parameters.

As the user moves to different locations, the level of interference and the distance between the user and co-channel BS's fluctuate. Due to this, fluctuations in SINR and BER are observed in modulated signals, including 64-QAM, 16-QAM, QPSK, and BPSK, employed for an adaptive modulation technique with SPM in Rician, Rayleigh, and AWGN channels. BER is used to assess the effectiveness of modulation techniques and wireless communication systems.

Comparison of SPM and commonly used equalizers

We will compare the SPM with the three most commonly encountered equalizers in communication systems: Linear Equalizers (LE), Decision Feedback Equalizers (DFE), and Adaptive Equalizers. The SPM is a simple model of signal spread, providing insight into free-space path loss and first-order environmental attenuation. Nevertheless, it depends on perfect conditions and fails to take into consideration intricate impairments, such as multipath interference, noise, and fading. LEs are used to counteract linear distortions, such as frequency-selective fading, where they compensate for the channel's frequency response. When there is low interference, they operate effectively; however, they do not perform well in complex settings, where distortions are experienced as multipath distortions or nonlinear fades. Adaptive equalizers adjust the channel parameters in real-time in response to variations in the channel. They are the best at operating variable channels in real-time, such as mobile situations. Adaptive equalizers can adapt to any signal by continually tuning to it, thereby providing better results than the SPM when the signal changes frequently due to environmental changes or user mobility. Nevertheless, the SPM can also be used under stationary conditions, same as the adaptive equalizers are computationally demanding; thus, they are not needed in situations that are more stable and where the channel has less probability of instability. The comparison highlights that the SPM is useful in providing foundational knowledge of signal propagation, but is limited to simpler, more repeatable scenarios where the conditions are not dynamic. It is known that equalization components (LE, DFE, and the adaptive equalizers) are effective in reducing the interference and distortion negative influences on the performance of a system. All equalizers have their purposes, and LE is used when there are no severe conditions, provided the multipath condition is met. In cases of dynamic and mobile conditions, the adaptive ones are far better. Table 4 illustrates the comparison of SPM with different equalizers in communication systems.

Parameters	Minimum Values	Optimized Values	Maximum Values
K1	10.5	20.5	25
K2	15	43	60
K3	−15	7.83	15
K4	0	0.5	0.9
K5	−8	−5.5	0
K6	−5	0	0
K7	−4	0	1

Table 2. Optimized values of calibrated SPM parameters.

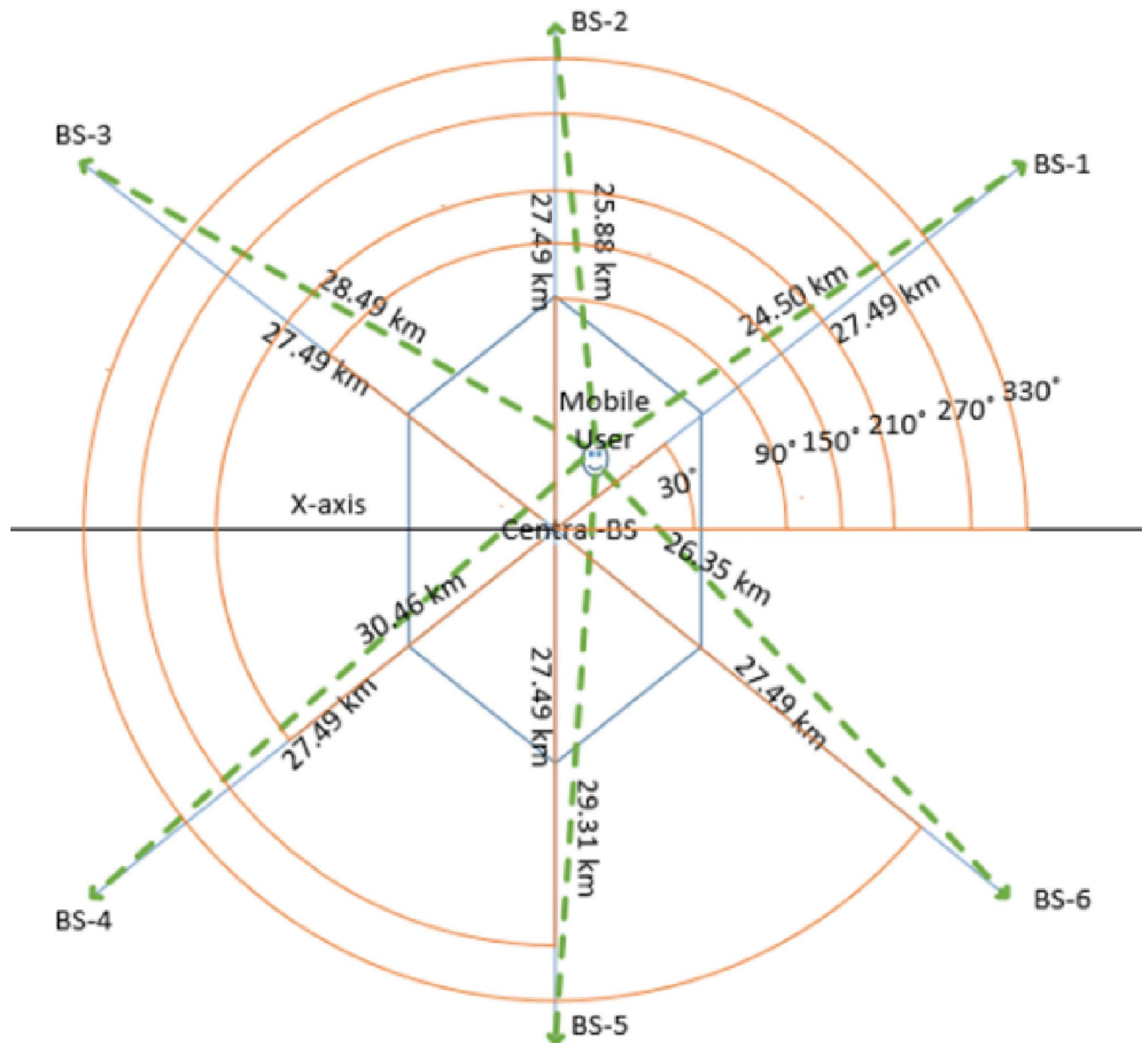


Fig. 3. Representation of distance and angle between co-channel mobile BSs and the primary BS, as well as the distance between the mobile user and the co-channel mobile BSs.

Parameters	Specification
Frequency	2×10^9 Hz
Transmitted power P_t	43 dBm
Modulation schemes	BPSK, QPSK, 16QAM, 64QAM
Channels	AWGN, Rayleigh, Rician
Radius of cell	6 km
Wavelength	0.15 m
Fixed User Mobility	0.1 km/h
Reference distance d_o	100 m
Traffic Model	$\lambda = 5$ packets per second
$G_T = G_R$	5dBi
Co-channel BSs	6
Distance between the user and central BS	3 km

Table 3. Simulation Parameters.

Equalizers	Advantage	Limitations
LE	In Low-Noise, it is very effective, Simple, and with Non-dispersive Channels	It is not effective in high Noise levels and with multi-path interference
DFE	Very effective to combine multi-path fading and ISI	The performance of DFE is very limited in high non-linear environments, and also complicated in implementations
Adaptive Equalizers	Dynamically adjusts to time-varying channel conditions; excellent for mobile environments	Computationally intensive; performance may degrade in stable or predictable environments
SPM	In ideal conditions, it provides a signal attenuation and a simple basic understanding of path loss	Does not account for multipath fading, noise, or dynamic channel conditions

Table 4. Comparison of SPM with different equalizers in communication Systems.

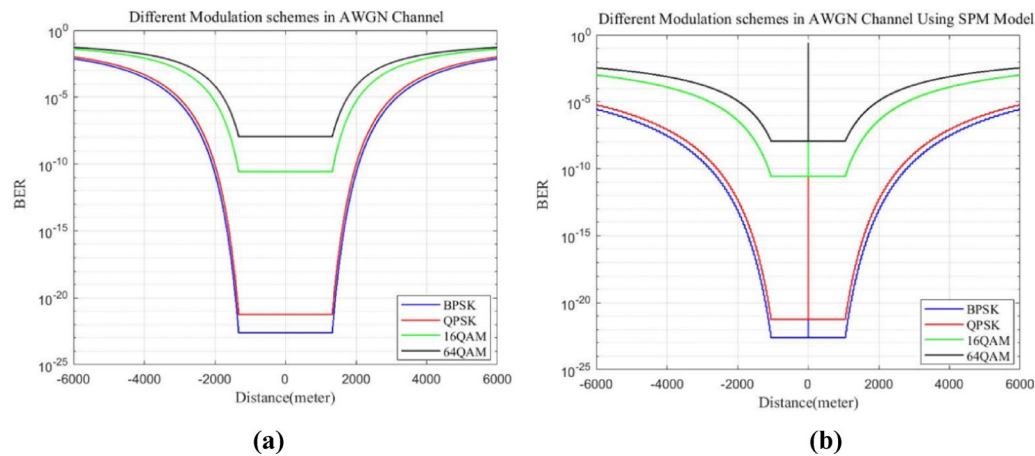


Fig. 4. The comparison of BER versus distance for different modulation schemes in the AWGN channel is provided in two scenarios: (a) without employing SPM, (b) with the employing of SPM.

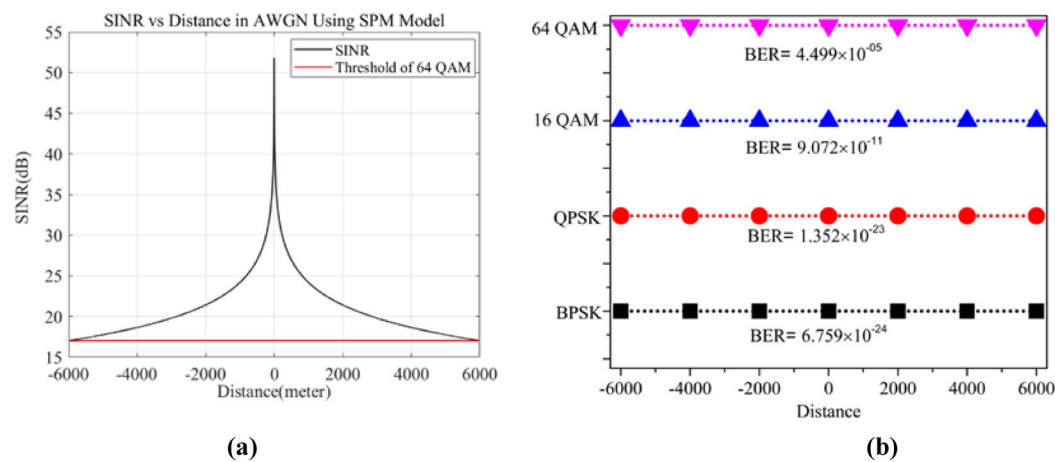


Fig. 5. (a) SINR versus Distance in AWGN channel employing SPM (b) BER of different Modulation Schemes at 6 Km in AWGN Channel employing SPM.

AWGN channel

Figure 4 demonstrates the direct relationship between the BER of different modulation techniques and the distance, whereas Fig. 5 displays that the SINR in the AWGN channel has an inverse relation with distance. Table 5 explains Figs. 4 and 5 by displaying the relation between BER, distance and SINR. As shown in Fig. 4, BPSK exhibits the lowest BER at lower data rates, whereas 64-QAM displays the highest BER at higher data rates in the AWGN channel. This disparity can be attributed to the simple constellation diagrams of low-order modulation schemes such as BPSK and QPSK. The constellation diagram of BPSK consists of two message points with a phase angle difference of 180° ⁴⁷, while QPSK has equidistant message points with a phase angle between symbols is 90° that are further apart. These characteristics make BPSK and QPSK more resistant to

Without Using SPM					
Distance (m)	SINR (dB)	BER			
		BPSK	QPSK	16-QAM	64-QAM
6000	7.299	4.471×10^{-03}	1.033×10^{-02}	1.701×10^{-02}	1.06×10^{-02}
5000	8.882	1.258×10^{-03}	3.335×10^{-03}	2.014×10^{-02}	2.226×10^{-02}
4000	10.82	2.893×10^{-04}	4.803×10^{-04}	8.378×10^{-03}	2.654×10^{-02}
3000	13.32	2.925×10^{-06}	7.88×10^{-06}	1.119×10^{-03}	3.749×10^{-03}
2000	16.84	2.247×10^{-11}	9.869×10^{-11}	5.065×10^{-06}	7.332×10^{-05}
1000	22.86	.*	.*	2.664×10^{-11}	1.144×10^{-08}
0	57	.*	.*	2.664×10^{-11}	1.144×10^{-08}
With Using SPM					
6000	17.02	2.925×10^{-06}	6.351×10^{-06}	1.008×10^{-03}	1.472×10^{-03}
5000	17.75	3.85×10^{-07}	9.427×10^{-07}	4.012×10^{-04}	1.167×10^{-03}
4000	18.65	3.417×10^{-08}	9.666×10^{-08}	1.344×10^{-04}	7.95×10^{-04}
3000	19.8	3.613×10^{-10}	1.343×10^{-09}	1.748×10^{-05}	1.802×10^{-04}
2000	21.42	8.717×10^{-14}	5.356×10^{-13}	4.301×10^{-07}	1.23×10^{-05}
1000	24.19	.*	.*	2.664×10^{-11}	1.144×10^{-08}
0	51.82	.*	.*	2.664×10^{-11}	1.144×10^{-08}

Table 5. BER vs. SINR and distance of various modulation schemes in AWGN channel with/without SPM. Note *: Indicated the situation where the BER is almost zero.

channel fluctuations, noise, distortion, attenuation, and fading, leading to a lower BER. BPSK, in particular, is less affected by the channel due to the considerable distance between its message points, resulting in a lower BER compared to QPSK. As a result, BPSK and QPSK are recommended for achieving acceptable BER levels in transmission scenarios characterized by poor link stability, extended transmission distances, and low SINR values.

Both 16-QAM and 64-QAM have complex constellation diagrams with closely spaced message points in the IQ plane, making them less immune to channel fluctuations. High-order modulation schemes like QAM are susceptible to interference, distortion, noise, and multipath fading, requiring a high SINR for accurate demodulation⁴⁸ and resulting in significant BER. Fading channels have a greater impact on the BER of a 64-QAM signal than a 16-QAM due to the close proximity of message points in the constellation diagram. 16-QAM and 64-QAM are suitable for favorable channel conditions and limited transmission ranges, minimizing deterioration.

The proposed technique using SPM in Fig. 4(b) improves the BER for all modulation schemes compared to the technique without SPM. With SPM, high-order bandwidth-efficient modulation schemes like 64-QAM achieve the target BER = 10^{-2} in the entire coverage area of the BS in an AWGN channel as compared to without using SPM (as shown in Fig. 4(a) and Table 5). SPM enables high data rate transmission across the entire 6 km cell coverage by improving SINR and reducing BER for 64 QAM, outperforming techniques without SPM. In fading channels, as the distance between the transmitter and receiver increases, the BER, attenuation, distortion, and noise increase while the SINR decreases. The adaptive modulation without SPM adjusts the modulation schemes based on distance and SINR thresholds to accomplish a desired BER = 10^{-2} , as shown in Fig. 4(a). However, the proposed technique using SPM achieves this target BER = 10^{-2} for 64-QAM across the entire distance range with high SINR, mitigating the effects of the wireless channel and maximizing system throughput, as shown in Fig. 4(b). Figure 4(b) demonstrates the effective utilization of all modulation schemes employing SPM for long-distance communication in the AWGN channel to meet the target BER = 10^{-2} . In wireless channels, BER and SINR change with distance. Short transmission distances result in low BER and high SINR, leading to high throughput. On the other hand, longer distances lead to higher BER, lower SINR, and reduced throughput.

Figure 5(a) illustrates the SINR and distance switching thresholds for 64-QAM with SPM, whereas Fig. 5(b) exhibits the BER of modulation schemes at a distance of 6 km in an AWGN channel with SPM. The inclusion of SPM with an adaptive modulation technique in the AWGN channel, as shown in Fig. 5(a), improves SINR (by 40%) for different modulation schemes, particularly in long-range distances, compared to the technique where only adaptive modulation is used.

Table 5 illustrates the BER of different modulation schemes under varying SINR and distance conditions in the AWGN channel. The 0 m distance stated in Table 5, derived from simulation results, serves as a theoretical reference point rather than representing the real-world situation of a co-located transmitter and receiver. At this distance, the SINR is high (51–57 dB), indicating an optimal condition with no path loss or interference, considering only AWGN. The reference SNR of 57 dB represents the highest possible signal strength at the beginning of the propagation channel, before distance-based losses occur. While the high SINR at 0 m is idealized and not feasible due to real-world limitations, it provides a baseline for understanding SINR degradation as distance and channel conditions change. The value of this high SINR at Distance = 0 m is determined by transmit power, antenna gain, and receiver noise figure. As the distance increases, these values are reduced gradually on the channel, as seen from the simulation results presented in Table 5. We believe that representing this 0 m reference SINR provides an overall sense of system behavior from ideal to gradual reduction with increased

Modulation schemes	Relative minimum distance	Data rate
BPSK	1	1
QPSK	$\frac{1}{\sqrt{2}}$	2
16-QAM	$\frac{1}{\sqrt{10}}$	4
64-QAM	$\frac{1}{\sqrt{42}}$	6

Table 6. Data rate and relative minimum distance between consecutive constellation message points regarding modulation schemes.

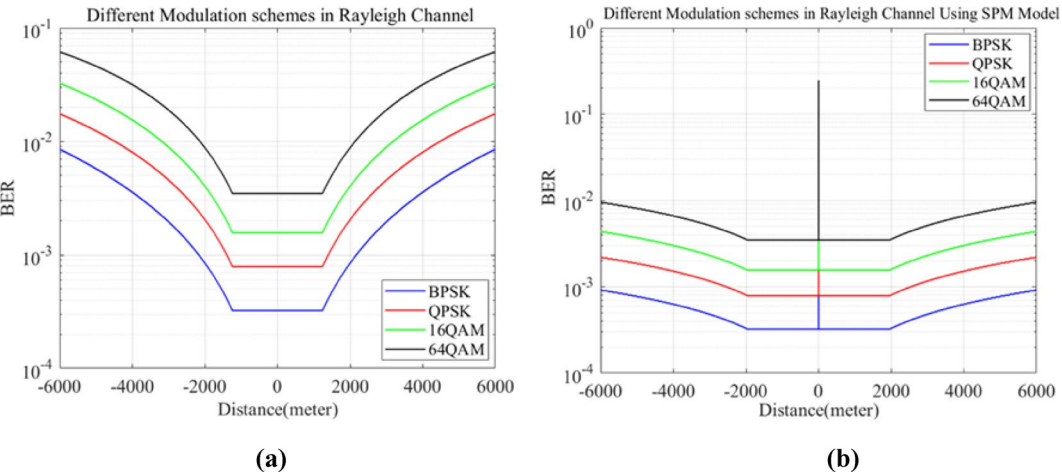


Fig. 6. BER versus distance of various modulation schemes in Rayleigh channel (a) without using SPM, (b) with using SPM.

distance. Each modulation type has a specific bit rate and a relative minimum distance between consecutive message points in its constellation diagram. Therefore, Table 6 provides information on the data rate and the minimum distance between successive symbols of BPSK, QPSK, 16-QAM, and 64-QAM.

In the AWGN channel, the absence of multipath phenomena results in path loss primarily determined by the distance separating the transmitter and receiver. There are many factors affecting the average transmission success probability between neighboring nodes, such as signal quality, antenna quality and interference⁴⁹. The implementation of adaptive modulation techniques improves the performance of modulation schemes in terms of BER through the utilization of various optimization parameters associated with SPM. Thus, the simple constellation diagrams, the low effect of the AWGN channel, as well as the proposed technique using SPM, are the main reasons for the lowest BER of BPSK and QPSK in the AWGN channel.

Rayleigh channel

Figure 6 shows that for modulation schemes like BPSK, QPSK, 16-QAM, and 64-QAM, the BER increases proportionally with distance, similar to the AWGN channel. Figure 7 illustrates that the SINR decreases inversely with distance in a Rayleigh fading channel. Table 7 reveals that the BER of various modulated signals is higher in the Rayleigh channel at lower SINR values compared to the AWGN channel for the same distances. Factors such as non-LOS transmission, diffraction, scattering, absorption, and reflection result in signal degradation, causing these signals to perform worse in Rayleigh channels than in AWGN. This is due to the multipath propagation of the signal in Rayleigh channels compared to the simple addition of noise in AWGN. Figure 6 (b) demonstrates that the adaptive modulation approach using SPM has a lower BER value compared to without using SPM in a Rayleigh fading channel (See Fig. 6(a)). Modulated signal characteristics like BER, SINR, path loss and interference fluctuate with changing distance, modifying the effects of multipath fading channels.

The approach employed with SPM favors high-order modulation signals such as 64-QAM in both large and small transmission ranges, achieving a target BER = 10⁻² with high SINR in a Rayleigh fading channel. This technique diminishes system complexity for all modulation schemes utilizing SPM by adjusting switching threshold values based on distance and SINR, thereby decreasing BER, interference, and mitigating the impact of factors like attenuation, noise, and distortion on modulated signals in a Rayleigh channel.

The switching threshold line of a higher modulation scheme such as 64-QAM using SPM is shown in Fig. 7(a). The Fig. 7 (a) shows that deploying SPM with an adaptive modulation technique in the Rayleigh fading channel improves the SINR (43%) of different modulation schemes, especially in long-range distances as compared to the technique, where only adaptive modulation is used. The modulation schemes exhibit elevated BER levels at a distance of 6 km in the Rayleigh channel, contrasting with their performance in AWGN conditions. Refer

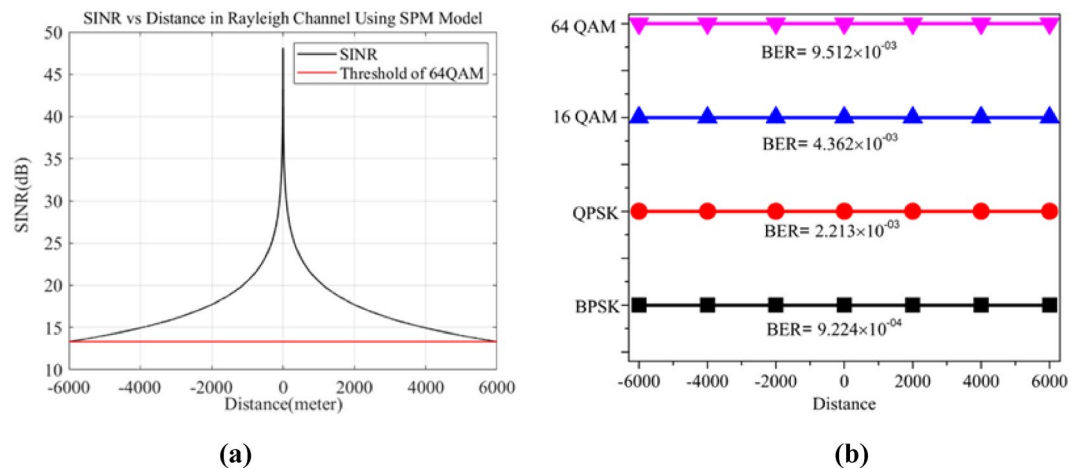


Fig. 7. (a) SINR versus Distance in Rayleigh channel employing SPM (b) BER of different Modulation Schemes at 6 Km in Rayleigh Channel employing SPM.

Without Using SPM					
Distance (m)	SINR (dB)	BER			
		BPSK	QPSK	16-QAM	64-QAM
6000	5.319	8.564×10^{-3}	1.756×10^{-2}	3.264×10^{-2}	6.156×10^{-2}
5000	6.903	5.733×10^{-3}	1.235×10^{-2}	2.339×10^{-2}	4.607×10^{-2}
4000	8.841	3.591×10^{-3}	8.078×10^{-3}	1.556×10^{-2}	3.187×10^{-2}
3000	11.34	1.966×10^{-3}	4.591×10^{-3}	8.963×10^{-3}	1.903×10^{-2}
2000	14.86	8.398×10^{-4}	2.02×10^{-3}	3.984×10^{-3}	8.706×10^{-3}
1000	20.88	3.232×10^{-4}	7.887×10^{-4}	1.564×10^{-3}	3.467×10^{-3}
0	53.32	3.232×10^{-4}	7.887×10^{-4}	1.564×10^{-3}	3.467×10^{-3}
With Using SPM					
6000	13.32	9.224×10^{-4}	2.213×10^{-3}	4.362×10^{-3}	9.512×10^{-3}
5000	14.05	7.648×10^{-4}	1.843×10^{-3}	3.638×10^{-3}	7.967×10^{-3}
4000	14.95	6.344×10^{-4}	1.534×10^{-3}	3.033×10^{-3}	6.665×10^{-3}
3000	16.1	4.797×10^{-4}	1.165×10^{-3}	2.306×10^{-3}	5.091×10^{-3}
2000	17.72	3.308×10^{-4}	8.07×10^{-4}	1.6×10^{-3}	3.546×10^{-3}
1000	20.49	3.232×10^{-4}	7.887×10^{-4}	1.564×10^{-3}	3.467×10^{-3}
0	48.12	3.232×10^{-4}	7.887×10^{-4}	1.564×10^{-3}	3.467×10^{-3}

Table 7. BER vs. SINR and distance of different modulation schemes in the Rayleigh channel with/without SPM.

to Fig. 7(b) for details in the Rayleigh channel and Fig. 5(b) for AWGN. This discrepancy is attributed to the significant distortion, attenuation, and path loss inherent in the multipath Rayleigh fading channel, which affects modulated signals and results in inferior performance of the Rayleigh channel compared to the AWGN channel.

Rician channel

In the Rician channel, the BER of various modulated signals is higher than in the AWGN channel, but lower than in the Rayleigh channel, according to Figs. 4 and 8, and 6, respectively. The reason for this phenomenon can be attributed to the characteristics of the Rician channel, which displays a SINR lower than that of the AWGN channel but higher than the Rayleigh channel, as detailed in Table 8. This discrepancy is a result of the Rician channel's composition of multiple weak multipath signals alongside at least one dominant LOS path. This minimizes Rician channel degradation, resulting in reduced signal distortion, attenuation, and noise. The BER remains constant in Rician channels at short distances, akin to Rayleigh channels due to the channel's minimal impact and the close proximity between mobile users and the BS. The technique employing SPM in Fig. 8 (b) accomplishes a constant BER at a significant distance of 2260 m, surpassing the performance without SPM at a distance of 1643 m in the Rician channel (see Fig. 8 (a)). The adaptive modulation schemes deploying SPM thus deliver lower BER values relative to distance than those without SPM in a Rician fading channel.

The simulation results show that the proposed technique, when used with SPM, performs better in Rayleigh, Rician, and AWGN wireless channels than when used without SPM or with a fixed modulation scheme. Fixed

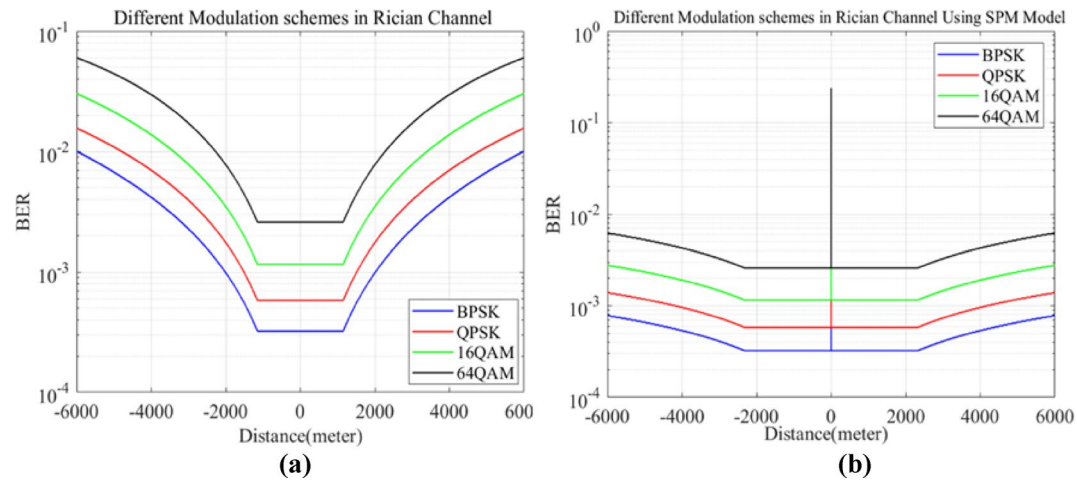


Fig. 8. BER vs. distance of different modulation schemes in Rician channel (a) without using SPM, (b) with using SPM.

Without Using SPM					
Distance (m)	SINR (dB)	BER			
		BPSK	QPSK	16-QAM	64-QAM
6000	5.899	9.971×10^{-3}	1.554×10^{-2}	3.007×10^{-2}	5.988×10^{-2}
5000	7.482	6.659×10^{-3}	1.079×10^{-2}	2.114×10^{-2}	4.396×10^{-2}
4000	9.421	4.159×10^{-3}	6.982×10^{-3}	1.379×10^{-2}	2.967×10^{-2}
3000	11.92	2.269×10^{-3}	3.93×10^{-3}	7.798×10^{-3}	1.718×10^{-2}
2000	15.42	9.897×10^{-4}	1.756×10^{-3}	3.491×10^{-3}	7.787×10^{-3}
1000	21.46	3.232×10^{-4}	5.817×10^{-4}	1.157×10^{-3}	2.589×10^{-3}
0	54.5	3.232×10^{-4}	5.817×10^{-4}	1.157×10^{-3}	2.589×10^{-3}
With Using SPM					
6000	14.02	6.965×10^{-4}	1.244×10^{-3}	2.473×10^{-3}	5.524×10^{-3}
5000	14.75	5.915×10^{-4}	1.058×10^{-3}	2.105×10^{-3}	4.705×10^{-3}
4000	15.65	4.797×10^{-4}	8.603×10^{-4}	1.711×10^{-3}	3.826×10^{-3}
3000	16.8	3.63×10^{-4}	6.526×10^{-4}	1.298×10^{-3}	2.904×10^{-3}
2000	18.42	3.232×10^{-4}	5.817×10^{-4}	1.157×10^{-3}	2.589×10^{-3}
1000	21.19	3.232×10^{-4}	5.817×10^{-4}	1.157×10^{-3}	2.589×10^{-3}
0	48.82	3.232×10^{-4}	5.817×10^{-4}	1.157×10^{-3}	2.589×10^{-3}

Table 8. BER versus SINR and distance of different modulation schemes in Rician channel with/without SPM.

modulation schemes, due to extreme attenuation and distortion, perform poorly in these channels, leading to significant errors in signal transmission and system degradation. Although effective within specific ranges of distances and SINR levels, these techniques are deemed inefficient in terms of bandwidth utilization due to their consistent use of the same modulation scheme under varying channel conditions.

Figure 9 (a) illustrates the 64-QAM switching threshold line in a Rician channel using SPM. Figure 9(a) shows that employing SPM with adaptive modulation in the Rician fading channel significantly enhances the SINR (by 41%) for different modulation schemes, particularly at long-range distances, compared to using only adaptive modulation. The BER values observed for all modulation schemes at a distance of 6 km in the Rician channel (refer to Fig. 9(b)) are superior to those in the Rayleigh channel (refer to Fig. 7(b)) but inferior to those in the AWGN channel (refer to Fig. 5(b)). The SINR and distance threshold values for the Rician channel are provided in Table 9, utilizing the technique with/without SPM to achieve a target BER = 10⁻². This approach serves to optimize the performance of the Rician channel by reducing BER, minimizing interference from neighboring mobile BS's, maximizing system throughput, and alleviating the channel's fading tendencies.

Table 8, presented below, shows the BER versus SINR and distance for various modulation schemes in a Rician channel, with and without using SPM. Table 9 presents the switching threshold values of SINR and distances for various modulation schemes (BPSK, QPSK, 16-QAM, and 64-QAM) used in the discrete adaptive modulation technique, with and without SPM, for AWGN, Rayleigh, and Rician fading channels, aiming to achieve a target BER = 10⁻². A condensed representation of simulation outcomes, with and without employing SPM, across AWGN, Rayleigh, and Rician channels is presented in Table 9. This table illustrates that the performance of the

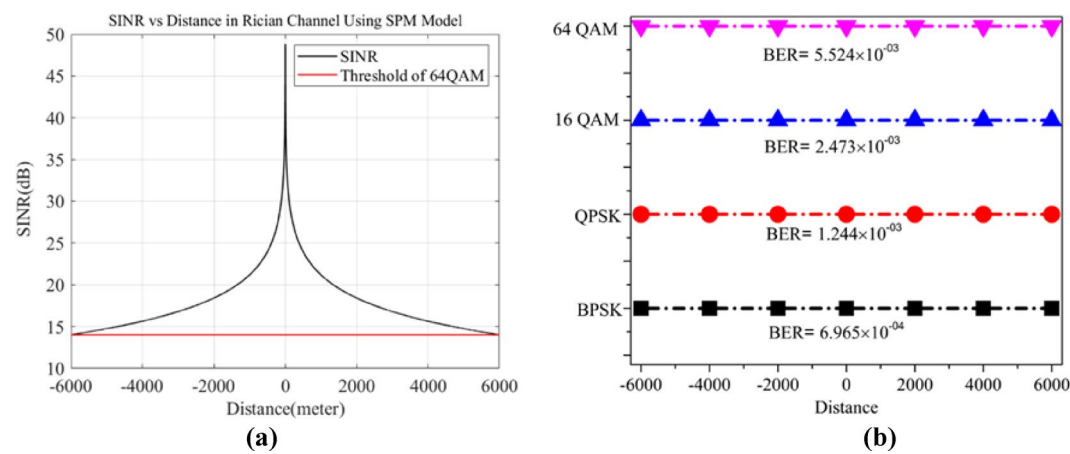


Fig. 9. (a) SINR vs. Distance in Rician channel using SPM (b) BER of various Modulation Schemes at 6 Km in Rician Channel using SPM.

Distance (m)			SINR (dB)			Modulation Schemes	Target BER
			Without Using SPM				
AWGN	Rayleigh	Rician	AWGN	Rayleigh	Rician		
6000 – 5948	6000 – 4483	6000 – 4779	7.29–7.87	5.319–7.851	5.899–7.875	BPSK	10 ^{–2}
5948 – 4115	4483 – 3174	4779 – 3383	7.87–10.48	7.851–10.85	7.875–10.88	QPSK	
4115 – 3543	3174 – 2145	3383 – 2261	10.48–14.58	10.85–14.25	10.88–14.38	16-QAM	
3543–0	2145–0	2261–0	≥ 14.58	≥ 14.25	≥ 14.38	64-QAM	
With Using SPM							
6000 to BS	6000 to BS	6000 to BS	17.02–51.82	13.32–48.12	14.02–48.82	64-QAM	10 ^{–2}

Table 9. The representation of SINR and distance switching threshold values for BPSK, QPSK, 16-QAM, and 64-QAM used in adaptive modulation technique with/without SPM in AWGN, rayleigh, and Rician channels.

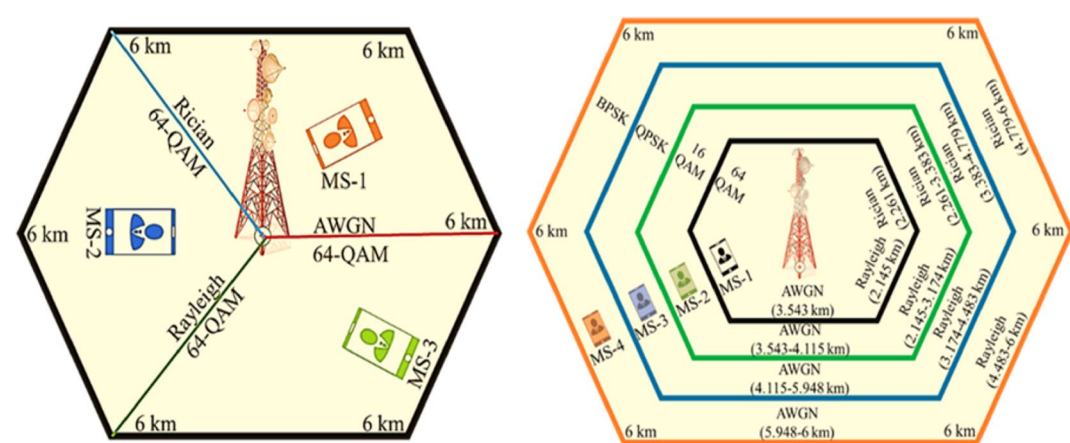


Fig. 10. Demonstration of distance-based selection of different modulation schemes in AWGN, Rayleigh and Rician channels (a) with and (b) without using SPM.

AWGN channel surpasses that of the Rayleigh and Rician channels, with the Rayleigh channel exhibiting inferior performance compared to the Rician channel in terms of BER and SINR at equivalent distance parameters.

Table 9 summarizes simulation results for different channels of a proposed technique with and without using SPM. The results show that deploying SPM with adaptive modulation increases the SINR by 40% in AWGN, 43% in Rayleigh fading, and 41% in Rician fading channels. This improvement allows all modulation schemes in the adaptive modulation technique to achieve the desired BER over a 6 km coverage area. The use of SPM enables the transmission of high data rates using 64-QAM signals in this coverage area. We can see from Figs. 4, 5, 6,

Previous work	Technique	Modulation type	Channel	Distance	SINR	SNR	Outcomes in terms of considered performance criteria (BER)
S.Barua et al ⁵⁰ .	Adaptive Modulation	BPSK, QPSK, 16-QAM	Rayleigh	✗	✗	✓	10^{-2}
H. Oh et al ⁵¹ .	Adaptive Modulation	BPSK, QPSK, 16-QAM, 64-QAM	AWGN, Impulsive Noise	✗	✗	✓	10^{-2}
A. M. Juliet et al ⁵² .	Adaptive Modulation	BPSK, QPSK, 16-QAM 64-QAM	AWGN	✗	✗	✓	10^{-4}
S.S. Hadi et al ⁵³ .	Adaptive Modulation	QPSK, 16QAM, 64QAM	AWGN, Fading channel	✗	✗	✓	0.11239
M.A. Azza et al ⁵⁴ .	Adaptive Modulation and Coding	QPSK, 16-QAM, 64-QAM	AWGN	✗	✗	✓	10^{-4}
Proposed work	Adaptive Modulation Using SPM	BPSK, QPSK, 16-QAM, 64-QAM	AWGN, Rayleigh, Rician	✓	✓	✗	AWGN= 10^{-11} Rayleigh and Rician fading $\approx 10^{-3}$

Table 10. Comparison table between proposed work and the state-of-the-art.

7, 8, 9 and 10; Tables 5, 7 and 8, and 9 that the adaptive modulation technique with SPM works better than the technique without SPM for BER, SINR, and high data rate signals over long distances. We achieve this enhanced performance by optimizing different parameters of SPM in the adaptive modulation technique.

Fig. 10 shows the coverage areas of different modulation schemes, deployed with and without SPM, in various wireless communication channels. The coverage areas of BPSK, QPSK, 16-QAM, and 64-QAM modulation schemes are analyzed without SPM, while specifically considering the 64-QAM scheme with SPM in AWGN, Rayleigh, and Rician channels to achieve a BER = 10^{-2} , as depicted in Fig. 10(b) and 10(a), respectively. Modulation schemes with higher information rates, such as 64-QAM and 16-QAM, exhibit enhanced bandwidth efficiency but suffer from degraded system performance in terms of BER at longer distances and lower SINR levels. Consequently, these high-order modulation schemes are recommended for use in scenarios characterized by shorter transmission ranges, minimal fading, elevated SINR levels, and favorable channel conditions, as illustrated in Fig. 10(b) when employing techniques without SPM. Conversely, lower information rate schemes, such as BPSK and QPSK, are preferable for extended transmission distances, reduced SINR levels, increased fading, and unfavourable channel conditions due to their superior performance compared to 64-QAM and 16-QAM, resulting in fewer errors under such circumstances, as shown in Fig. 10(b). The incorporation of SPM enables the utilization of 64-QAM across the entire coverage area of AWGN, Rayleigh, and Rician channels, as shown in Fig. 10(a), facilitating the attainment of the target BER. Thus, the proposed approach, employing SPM, represents an optimal strategy for error reduction, throughput maximization, and accurate signal transmission over expansive radio coverage areas compared to the technique that excludes SPM. Table 10 provides a comparison of our proposed work with previous cutting-edge methods. As may be observed, the proposed work in this research is better than other people proposed approaches

Conclusions

This paper focuses on addressing multipath fading losses over a 6,000-meter distance in AWGN, Rayleigh, and Rician channels within a cellular communication system, targeting parameters such as BER and SINR. The proposed technique enhances the performance of modulation scheme such as 64-QAM in terms of (BER = 10^{-2}), and SINR for different communication channels including AWGN (40%), Rayleigh (43%), and Rician (41%). The results mentioned above are achieved by optimizing the different parameters of SPM, which enhances the overall performance of adaptive modulation based on SINR and distance (a 6000-meter coverage distance of the BS) in the three channels mentioned above, compared to the technique without using SPM and fixed modulation schemes. The impact of CCI from six neighbouring BS's on a mobile user situated 3 Km away from the central BS is examined. The CCI is influenced by factors like the number and spacing of co-channel BS's, as well as the location of the mobile user, thereby affecting performance metrics like BER and SINR in a wireless communication system. An adaptive approach utilizing SPM is proposed to achieve a target BER = 10^{-2} for 64-QAM, ensuring high throughput, minimizing channel fading, mitigating CCI, and efficiently utilizing bandwidth, with reduced power consumption and improved radio coverage. Future research may explore the scalability of the proposed adaptive modulation method by evaluating it with more intricate channel models and diverse modulation schemes in real-world environmental situations, including variable weather conditions and urban density. Additionally, the application of machine learning methods may enhance parameter selection for further optimization, potentially leading to reduced computational overhead. Exploring cross-layer optimization techniques to mitigate issues such as packet loss and limited bandwidth in multi-user environments may enhance the effectiveness of the proposed system.

Data availability

The data used and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Each author in this article contributed their distinct expertise and responsibilities in a collaborative way. R.K. and Z.U.K. were primarily responsible for formulating the study's concepts and determining the research direction. R.K. provided methodological frameworks so that the study's approach was rigorous and coherent. N.A. assumed responsibility for the software implementation, which is critical for data analysis and interpretation. Z.U.K., S.T., and S.A. collaborated to validate the findings, ensuring the strength and reliability of the conclusions reached. S.U.K. supervised formal analytic techniques, while R.K. and J.K. carried out essential studies for data collection and interpretation. N.A. supervised the use of resources, while S.U.K. thoroughly examined the collected data. S.T. handled further modifications and editing after R.K. initially wrote the manuscript. S.A. used data visualization to improve the presentation of significant findings. S.U.K. and R.K. provided supervision throughout the study process, ensuring complete conformity to scholarly guidelines. S.U.K. and J.K. collaborated on project administration tasks, with N.A. leading the funding acquisition task. All authors provided input during the manuscript drafting stage.

Declarations

Competing interests

The authors declare no competing interests.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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